

ALY-6050 MOD 4 – Lab I

A Minimizing Decision Model

In this lab, we will first create a minimization model. Next, we will solve that model and will obtain a “best” solution. We will use R as the primary software for this lab.

Watch this video:

<https://www.youtube.com/watch?v=fKkzqk3RMLc&feature=youtu.be>

Scientists have determined that, during long flights, most species of birds select their speeds so that their total consumed energy during the flight is minimized. The bird's total energy spent (in calories per kilometer) during such flights has been modeled according to the following mathematical relationship:

$$\text{Total energy: } E = \frac{0.075(v-33)^2 + 20}{v} \quad \text{calories per kilometer}$$

In this model, the variable v represents the bird's speed (or velocity) during the flight in kilometers per hour.

Task:

Determine the particular value of the bird's speed that minimizes the total energy spent by the bird per kilometer of its flight.